

Problem 1

Planck's function can be encountered in several forms depending on the conventions and preferences of different scientific fields. The various forms of the law for spectral radiance are summarized below.

Frequency ν	$B_\nu(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_B T)} - 1}$
Wavelength λ	$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_B T)} - 1}$
Wavenumber $\tilde{\nu}$	$B_{\tilde{\nu}}(\tilde{\nu}, T) = 2hc^2 \tilde{\nu}^3 \frac{1}{e^{hc\tilde{\nu}/(k_B T)} - 1}$

The conversion between different formats is based on two relations. One is the relation between frequency ν , wavelength λ , wavenumber $\tilde{\nu}$ and speed of light c , which is used to convert the fractional bandwidth, for example, $x = \frac{h\nu}{k_B T} = \frac{hc}{\lambda k_B T}$ because $c = \lambda\nu$. The other relation is the energy conservation, i.e., $B_\nu d\nu = B_\lambda d\lambda = B_{\tilde{\nu}} d\tilde{\nu}$ (see textbook section 1.2.1 for detail)

- 1) Starting with the Planck's function in terms of wavelength λ , derive the format for Frequency and Wavenumber.
- 2) Derive the inverse of the Planck's function B_λ and $B_{\tilde{\nu}}$, which is often called the “brightness temperature” ($BT = B^{-1}(\lambda, I)$) that corresponds to the emission of a blackbody at a wavelength and wavenumber

Problem2:

- 1) An infrared scanning radiometer aboard a meteorological satellite measures the outgoing radiation emitted from the earth's surface in the $10\text{ }\mu\text{m}$ window region. Assuming that the effect of the atmosphere between the satellite and the surface can be neglected, what would be the temperature of the surface if the observed radiance at $10\text{ }\mu\text{m}$ is $9.8\text{ W m}^{-2}\text{ }\mu\text{m}^{-1}\text{ sr}^{-1}$?
- 2) A black land surface with a temperature of 15° C emits radiation at all frequencies. What would be the emitted radiances at $0.7\text{ }\mu\text{m}$, 1000 cm^{-1} , and 31.4 GHz ? Use appropriate Planck functions in the calculations.

Problem 3:

1. What are the wavelengths (in the unit of cm) and wavenumbers (in the unit of m^{-1}) of C-band and W-band radars with the frequencies of 6GHz and 98 GHz, respectively?
2. What is the energy in terms of electronvolt (eV) of a photon from a red light (wavelength 650 nm)?
3. What is the wavenumber corresponding to the “atmospheric window” in the wavelength range of 8 and 12 μm ?

Problem 4:

Venus's orbit has a radius of about 0.72 Astronomical Unit. The planetary albedo of Venus is about 0.77. Compute the Equivalent Blackbody Temperature (T_E) of Venus based on energy balance equation. Venus has an atmosphere of thick clouds and 90% CO_2 and has a surface temperature $T_s=750\text{K}$. Explain the difference between T_E and T_s

Problem 5:

The intensity for collimated sunlight is usually written as $F_0 \delta(\mu - \mu_0) \delta(\phi - \phi_0)$, where F_0 is the solar irradiance, μ_0 is the cosine of the solar zenith angle, ϕ_0 is the solar azimuth angle.

For simplicity, assume $F_0 = 1$, $\mu_0 = 0.5$. This sunlight is illuminating on a Lambertian reflector with an albedo of $\rho = 0.3$. The reflected light by a Lambertian reflector is *isotropic*. The albedo of a Lambertian reflector is defined as:

$$\rho = \frac{F \uparrow}{F \downarrow}, \quad (1)$$

where $F \downarrow$ is the incoming flux and $F \uparrow$ is the reflected flux.

Derive:

- 1) The downward flux and upward flux at the surface of the Lambertian reflector.
- 2) The actinic flux at the surface of the Lambertian reflector.

Problem 6:

A satellite orbiting around the Earth is heated by solar radiation and by the operation of the instruments it carries. The only way to control a satellite's temperature in outer space is radiative cooling. Let the instruments release heat of a power of 200W. Assume the satellite to be a 1 m^3 cube with one side of the cube oriented perpendicular to the Sun.

- 1) Calculate the emissivity for thermal radiation of the satellite housing required to establish a thermal equilibrium at a temperature of 10°C if the satellite surface has a solar albedo of 0.2 or 0.8.
- 2) How does the satellite temperature change if you take into account the thermal emissions by the Earth? Assume the Earth's and its atmosphere's effective emission temperature to be -31°C and the satellite to orbit at 300 km above the Earth's surface such that 1 m^2 of the satellite receives the thermal radiation from the Earth. Use the IR emissivity values from the last question.